

Neato D7 - Fan Stuck... but fan is fine. (hack fixed posted)

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Neato D7 - Fan Stuck... but fan is fine. (hack fixed posted)

by **jdredd** » December 31st, 2019, 5:11 pm

My Neato D7 promptly quit working after latest 4.5.3 firmware update. Not sure if firmware or just my luck. Literally the day I did the update then tried to run it, I get a fan stuck error.

I am pretty comfy working on these as I've had the D7 and two D3's (one on each floor) for nearly two years.

I took the unit apart, and manually powered the fan with 12v (its 14.4v fan) and works fine.

Hooked it to an Arduino even to read the yellow signal wire and do some PWM control via the blue wire.

I used the USB > Serial interface to play around to turn some things off an on and what not but not able to do anything with the fan. Not sure if goofing up the commands, or if it works and just no power control from the logic board is just happening.

Apparently can't go back in firmware updates... and no way to force an update to reflash again?

Any commands via the USB > Serial I can use to toggle fan off and on?

Anything on the logic board itself I can check out?

Last edited by **jdredd** on January 5th, 2020, 11:56 pm, edited 2 times in total.

NeatoToolio

Windows @ <https://github.com/jdredd87/NeatoToolio ... Toolio.exe>

Google Play Store @ <https://play.google.com/store/apps/deta ... eatoToolio>



jdredd
Robot Addict

Posts: 181
Joined: December 31st, 2019, 4:57 pm

Re: Neato D7 - Fan Stuck... but fan is fine.

by **glnc222** » December 31st, 2019, 9:13 pm

The Neato USB text command manual top of forum. Commands can be issued with a terminal emulator or with Neato Control, in the command option there, enhanced with logging features and repetition.

Test Mode must be set to operate motors: TestMode ON, TestMode OFF

and turned off manually before disconnecting (proper exit from Neato Control might do this).

To activate the fan I think it is SetMotor VacuumOn / VacuumOff, with an additional per cent speed parameter if wanted; I recall default is 90 per cent (I think this is actually the duty cycle of the PWM signal).

The motor appeared in tests to be smart, in that the PWM input sets RPM maintained internally by the motor. There is also a tachometer output, which would be used by the software to detect stuck conditions. Apparently no adjustment of the PWM is needed to maintain RPM.

So stuck fan conditions would involve something with the tach signal or its interface or processing by the cpu. The signal would appear used internally in the motor to maintain RPM itself. A signal driver for interface might be defective.

These tests done on old XV fans, with a more efficient fan used in the Botvac, for its wider intake.

A multimeter could have frequency and duty cycle scales for measuring the tach signal, without oscilloscope.

The fan current reported over USB appeared incorrect in the original Botvac with the shunt incorrectly placed; it seemed to measure only a small control current instead of the fan load around 1 amp. Total bot current reported can be used to get the actual fan load. With fixed RPM the fan load varies with the airflow which can be approximated with this measure, varied according to clogging of the filter. See thread Electronic Filter Monitoring.



glnc222
Robot Master

Posts: 5491
Joined: January 23rd, 2012, 8:19 pm
Location: The Villages, Florida
Contact: 

Re: Neato D7 - Fan Stuck... but fan is fine.

by **jdredd** » January 1st, 2020, 2:10 pm

Ok.. So the manual that is out there is the latest.... Since this was a Dx model ...

I looked at Neato Control and found the source ... Uses Delphi.. (which i use for a living for a living)

Sadly the fan itself is good as I can power and control it via PWM and read the signal via an Arduino, so assume either firmware fracked up or just my luck something on the motherboard crapped out literally right after the firmware update to no longer power/control the fan anymore. In my case the D7 doesn't even spin the fan at all.



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Robot Addict

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To humor myself I ordered a replacement fan... if its the logic board, probably sit on it for a while to find a board for "cheap" or keep it around for spare parts for the two D3's I also use in the house.

Terminal wise, pretty straight forward on most things.. may keep playing with that... and if time allows, may end up doing my own test app, as was not a fan of the neato control... and do Windows/Android flavor with Delphi.. as I can do the USB to Serial control in android... 😊

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Re: Neato D7 - Fan Stuck... but fan is fine.

by **glnc222** » January 1st, 2020, 6:57 pm

Alternative interface software in Python for Linux and Windows, thread <http://www.robotreviews.com/chat/viewto ... 20&t=19708>

So the system board is not generating the PWM signal when turned on with USB command? The power lines go on when the bot is cleaning or when TestMode is on, but the fan does not turn until the PWM is on (or at least the main brush). This allows the power line to be used for an added side brush motor on the old XV.

Re: Neato D7 - Fan Stuck... but fan is fine.

by **glnc222** » January 1st, 2020, 7:22 pm

Neato Control is from a Russian programmer as a hobby. The environment used was changed to Delphi at some early time, I forget the original. I think the source code is included in the download package. One thing in it is the USB device ID's for XV and Botvac, not sure about WiFi models, so it can find the port number automatically. Otherwise use Windows utilities to find the port number when connected. The ID is then available in Windows advanced options in device mgr or devices&printers.

Re: Neato D7 - Fan Stuck... but fan is fine.

by **jdredd** » January 1st, 2020, 10:53 pm

Well, I hooked up one my D3's and sent "SetMotor VacuumOn" and fan kicked on just fine.

So I take the the logic board is bad in the D7, as I swapped fans and the D7 didn't kick the fan on. 😊

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Re: Neato D7 - Fan Stuck... but fan is fine.

by **jdredd** » January 2nd, 2020, 1:53 am

Had a little bit of time to play around....

Writing object wrappers for each command... so far.. GetCharger is my first 😊

[media]_cT94HzRh-Y[/media]

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Re: Neato D7 - Fan Stuck... but fan is fine.

by **jdredd** » January 2nd, 2020, 1:54 am

Had a little bit of time to play around....

Writing object wrappers for each command... so far.. GetCharger is my first 😊

Nothing special but it does "work" 😊

[media]_cT94HzRh-Y[/media]

Basically connect... get some results every ~1 seconds.

Unplugged and plugged charging dock station while running.

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Re: Neato D7 - Fan Stuck... but fan is fine.

by [vic7767](#) » January 2nd, 2020, 10:57 am



vic7767
Robot Master

Posts: [15556](#)
Joined: January 14th, 2006, 7:31 pm
Location: Houghton Louisiana - USA

Re: Neato D7 - Fan Stuck... but fan is fine.

by [jdredd](#) » January 3rd, 2020, 3:34 pm

Doing tests, I got an old case fan to run... then swapped over to the D7 fan..

Not sure but started to smell "burning" of some sort.... then no fan power anymore again.

So something on the logic board must be toast that supplies power to the 4 pin power plug 🍷

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jdredd
Robot Addict

Posts: [181](#)
Joined: December 31st, 2019, 4:57 pm

Re: Neato D7 - Fan Stuck... but fan is fine. (hack fixed pos

by [jdredd](#) » January 6th, 2020, 12:01 am

I got my D7 working via a simple hack.

4pin fan power connector cut off and some female pin plugs added.

Red (power)
Black (ground)
Yellow (pwm)
Blue (signal)

I skip connecting the yellow wire. With this plugged in, the fan doesn't work. Assume whatever controls PWM from the logic board is stuck commanding 0 % always.

I also tap the ground wire now to also get ground from the that back side of the CAP in the video, near the fan connector. Using just the ground off the 4 pin fan connector, does not seem to be enough.

So using ground also off the CAP, this more or less gives the fan an ON or OFF mode. No throttle anymore.

Once energized, fan is gonna be on. So PWM can no longer power the fan but then also command it to 0 % duty cycle.

I can live with this.

Ran it a number of times, and so far it works!

Only down side I can tell so far, is it doesn't power off fan when it doesn't need really need it running. But really don't care either. Probably shorten the life. But since these are high end Computer server fans that are ment to run 24/7 I am sure the life span will be just fine.

Test video on my table... @ <https://www.youtube.com/watch?v=F1HpmwjaPdk>

Test video of the D7 back together and doing a SPOT pass @ https://www.youtube.com/watch?v=5cPs_9xFuaU

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masterx81
Robot Addict

Posts: [134](#)
Joined: May 9th, 2019, 5:08 am

Re: Neato D7 - Fan Stuck... but fan is fine. (hack fixed pos

by [jdredd](#) » January 6th, 2020, 9:08 am

“ masterx81 wrote:

I think that q3 is the transistor that drive the blower. Seem a k38 marking code. Seem a bss138. You can check if with a scope you get the pwm signal at the gate, maybe he is faulty. I think that leaving the fan always on reduce battery life



jdredd
Robot Addict

Posts: [181](#)
Joined: December 31st, 2019, 4:57 pm

You are most likely right. I keep calling it a cap...

Once I have it apart again soon, may get some closer pictures and probe around more.

I am sure PWM is dead....

Also I am sure ground is goofed in some way... as just using the power and ground pins alone on the fan connector do not work... but tapping off the back side of that transistor? for ground, seems to give it what it needs.

As for battery life... luckily its a new ion battery... plus can recharge and continue if need be 😊

It ran last night and then scheduled this morning, did my whole first floor in one pass.. which is only ~850 square feet of area.

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Re: Neato D7 - Fan Stuck... but fan is fine. (hack fixed pos

by masterx81 » January 6th, 2020, 9:23 am

I think that there must be a +vcc fixed, one ground (via shunt resistor and q3 transistor), then a pwm signal. The pwm signal must come some from other chip, need to check on the board.

masterx81
Robot Addict

Posts: 134
Joined: May 9th, 2019, 5:08 am

Re: Neato D7 - Fan Stuck... but fan is fine. (hack fixed pos

by jdredd » January 6th, 2020, 10:11 am

Yah. Fan pin 1, gets voltage.

Here is a video I made earlier yesterday, to test against a small test fan.

Can see how using the power/ground pins off the fan connector, just wiggles it.

<https://www.digikey.com/product-detail/ ... &pmt=&pkw=>

E3P 100 35V , which shows to be a Capacitor... not a transistor ... which with the orientation on the board, back side, that is the ground..

So I think 2 issues...

- 1) Ground pin, where ever that is leading... is weak... very weak...
- 2) PWM is dead. So 0 PWM signal... causes fan to think its being commanded to be off.

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Re: Neato D7 - Fan Stuck... but fan is fine. (hack fixed pos

by masterx81 » January 6th, 2020, 4:49 pm

Uhm pasted wrong link?

masterx81
Robot Addict

Posts: 134
Joined: May 9th, 2019, 5:08 am

Re: Neato D7 - Fan Stuck... but fan is fine. (hack fixed pos

by jdredd » January 6th, 2020, 5:48 pm

masterx81 wrote:
Uhm pasted wrong link?

jdredd
Robot Addict

Posts: 181
Joined: December 31st, 2019, 4:57 pm

Right link.... but me mis-understanding what you were getting at.

The link was for the CAP I am getting the new ground for, to run for the fan as the ground pin on the fan connector doesn't work very well at all. Thought you were talking about that.....

Q3 / K38 / BSS138, that part you mentioned, not 100% sure what exact component on the board that is currently (as i dont have the board in front of me)

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Re: Neato D7 - Fan Stuck... but fan is fine. (hack fixed pos

by masterx81 » January 6th, 2020, 6:06 pm

masterx81
Robot Addict

Posts: 134

Re: Neato D7 - Fan Stuck... but fan is fine. (hack fixed pos

by **jdredd** » January 11th, 2020, 10:46 pm

Quick update...

Moved away from what I had... it worked, but does run fan at 100% cranking speed...

I now have a bit better setup going.

Using a buck converter, i can scale back the voltage. Neato doesn't run the fan at full 100% cranking speed it looks anyways by default.

I tap off the solder point for the battery ground pin.

Tap power off the fan connector power pin.

Signal wire from fan, to fan connector power pin.

No PWM wire connected... (commands 0 all the time)

This works pretty well... only problem I have is the buck converter multi-pack I got for the D3 side brush hack, wasn't ment for such an AMP pulls... so i need to scale it down to around 6000-7000rpm.

I'll have a much better one tomorrow in the mail that can take 5 amps... then will see how that goes to get a few more extra RPM going.

So far this has worked well.. done 4 passes on the 1st floor. Does each pass on the 1st charge. With the previous deal, ending up starting to take 2 charges to complete.

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Re: Neato D7 - Fan Stuck... but fan is fine. (hack fixed pos

by **jdredd** » January 12th, 2020, 8:27 pm

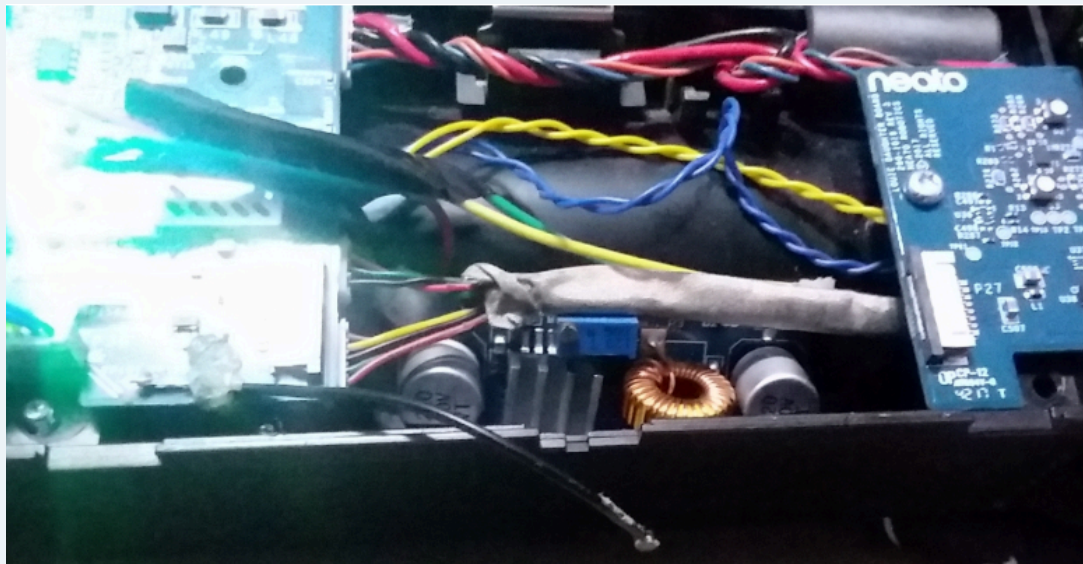
Hopefully last update....

Upgraded to a proper buck converter for the pull the fan needs.

Use one from @ <https://www.amazon.com/gp/product/B00LO...UTF8&psc=1>

Video @ https://www.youtube.com/watch?v=odQp_tm-y9c

Fits nicely in the area here.. kinda of tilted in there.



Sooooo.... hopefully as mentioned, last I'll have to deal with this 😊

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